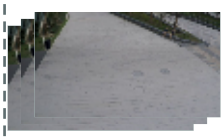
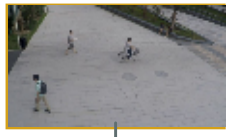
INPUT VIDEO



every 16th frame*



current frame f_t

❄ EVERY MODEL FROZEN · NOTHING TRAINED

No parameter is updated at any stage.
The 59 phrases were fixed before scoring, and
the same lists are used for every camera.

P · IMAGE BRANCH

❄ **PE-Core image encoder**

PE-Core-L-14-336 · 336 × 336 px

$v_t \in \mathbb{R}^{1024}$

unit-norm frame vector

TEXT BRANCH · 59 FIXED PHRASES

NORMAL SIDE (11)

"a calm and safe public area" + 2 more
C: 8 calm phrases
"an empty walkway"
"a person standing still"
"two people talking calmly"
... no look-alike actions

ABNORMAL SIDE (48)

B: 12 types × 4 actions
"two people punching each other"
"a cyclist riding through a crowd of pedestrians"
"a car on fire"
... written once, frozen

❄ **PE-Core text encoder, averaged per side**

FRAME SCORING

prototypes: each side's unit vectors averaged, then renormalised

e^+ **normal**

e^- **abnormal**

abnormal similarity minus normal similarity

$$s_t = \sigma(\lambda(\langle v_t, e^- \rangle - \langle v_t, e^+ \rangle))$$

$\lambda = 100$, fixed · σ = sigmoid

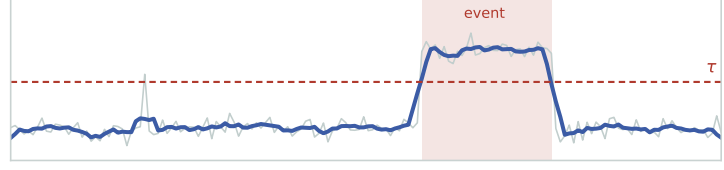
TEMPORAL SMOOTHING

centred moving average

$$\tilde{s}_t = \frac{1}{|\mathcal{W}_t|} \sum_{k \in \mathcal{W}_t} s_k$$

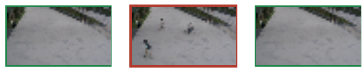
5 sampled frames* · raw scores

illustration: raw s_t (grey) vs smoothed $\tilde{s}_t$ (blue)



OUTPUT

flag if $\tilde{s}_t \geq \tau$



anomaly score for every frame

REMOVED FROM THE EARLIER DESIGN

· no scene sentence written for each camera
· no captioning model (LLaVA) at any stage
· nothing that depends on the first frames

A new camera needs no new data: the same 59 phrases · no labels · no gradients

* ShanghaiTech follows its benchmark protocol: every 2nd frame, a window of 31, and scores rescaled within each clip